AWS – INTRODUCTION

Class no.1

Assignment: Traditional vs. Modern IT & AWS Cloud Infrastructure

Traditional vs. Modern IT

1. Traditional IT (On-Premises)

- **Infrastructure Location:** Servers and hardware are hosted *on company premises*.
- **Maintenance Cost:** High upfront capital expenditure (CAPEX) for hardware, plus ongoing costs for upgrades, electricity, cooling, and IT staff.
- **Security Problems:** Greater control but vulnerable to physical damage, insider threats, and outdated patching if not maintained properly.
- **Communication:** Access typically through local networks or VPNs; limited scalability for remote collaboration.

2. Modern IT (Off-Premises / Cloud)

- **Infrastructure Location:** Hosted *off-premises* by cloud providers.
- **Maintenance Cost:** Lower upfront cost, pay-as-you-go model (OPEX). Providers handle upgrades and scaling.
- **Security Problems:** Providers invest heavily in security, but risks include misconfigurations, shared responsibility, and compliance challenges.
- **Communication:** Seamless global access via the internet; supports remote work, collaboration tools, and APIs.

Difference between Traditional and Modern IT :

Aspect	**Traditional IT (On-Premises)**	**Modern IT (Cloud)**
Location	Company premises	Provider’s data centers
Cost	High CAPEX + maintenance	OPEX, scalable
Security	Full control, local risks	Shared responsibility, advanced tools
Communication	Local/VPN	Internet-based, global

Modern IT – Cloud Computing & AWS

1. Cloud Service Providers

- **Top providers (2025–26):** Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), IBM Cloud, Oracle Cloud, Alibaba Cloud, Salesforce, SAP, Tencent Cloud, Huawei Cloud.

2. What is Cloud Computing?

- Delivery of computing services (servers, storage, databases, networking, software, analytics, AI) over the internet.
- **Working with Cloud Services:** Users access resources on demand, scale instantly, and pay only for usage.

3. Why AWS?

- **Largest market share** and most mature ecosystem.
- Offers **397+ services** (2026), including compute (EC2), storage (S3), databases (RDS), AI/ML, IoT, and security.

4. AWS Regions & Availability Zones

- **Global footprint (2026):** 39 Regions, 123 Availability Zones (AZs).
- **India:**
  - **Mumbai (ap-south-1):** 3 AZs (ap-south-1a, ap-south-1b, ap-south-1c).
  - **Hyderabad (ap-south-2):** 3 AZs.

5. Purpose of Regions & AZs

- **Regions:** Geographically isolated areas to meet compliance and latency needs.
- **Availability Zones:** Independent data centers within a region, designed for fault tolerance and high availability.

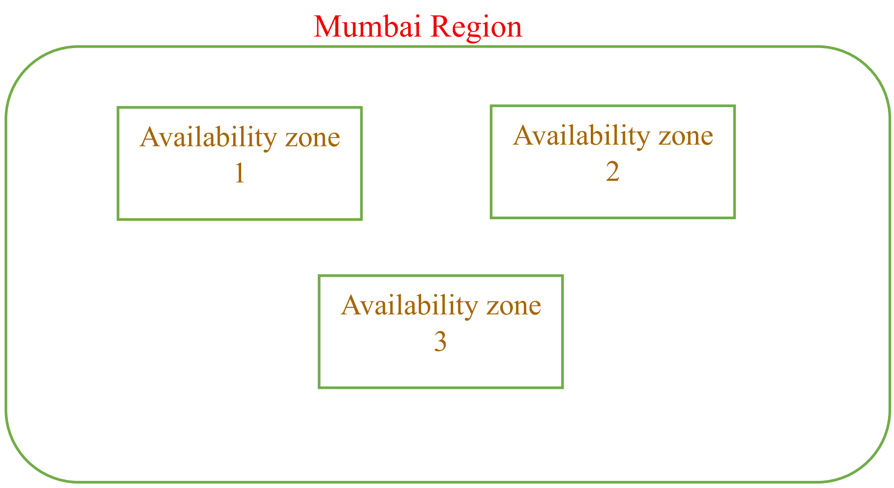

Figure 1.a : Regions and zones visual representation